"On my honor, I have neither given nor recieved any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code violation of these principles will lead zero to grade and is subject to honor discipline issues"

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Q1

$$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$ → We clearly see that This elimination matrix makes the number under the first pivot zero.

So $3\alpha + (-8)\cdot(1) = 0$ and $2 + (m)(1) = 0$

$3\alpha = 8$

$\boxed{\alpha = \frac{8}{3}}$

$\boxed{m = -2}$

So, this is the first elimination matrix

$$E_1 \cdot A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & -8\alpha+5 & -\gamma & -8\alpha+1 \\ m+2 & m\alpha+\beta & -m\gamma+3 & \alpha^2+m\alpha \\ 0 & -\alpha+14 & \gamma-6 & -\alpha+2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$ we clearly see that this is the second elimination matrix

$$= \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & \frac{3\beta-16}{3} & 2\gamma+3 & 16/9 \\ 0 & 34/3 & \gamma-6 & -2/3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & \frac{3\beta-16}{3} & 2\gamma+3 & 16/9 \\ 0 & 34/3 & \gamma-6 & -2/3 \end{bmatrix}$$

E_2 $E_1 \cdot A$

So $b + 11 = 0$ $\quad \boxed{b = -11}$ $\qquad \frac{-49k+34}{3} = 0$ $\quad \boxed{k = \frac{34}{49}}$

$$\Rightarrow \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -\frac{49}{3} & -\gamma & -61/3 \\ 0 & b+11 & 3\gamma+3 & 199/9 \\ 0 & \frac{-49k+34}{3} & -\gamma-k\gamma-6 & \frac{-61k-2}{3} \end{bmatrix} = \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & 0 & 3\gamma+3 & 199/9 \\ 0 & 0 & \frac{15\gamma-294}{49} & -14,775 \end{bmatrix}$$

$E_2 E_1 A$

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Then $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$ this is the third elimination matrix that makes the number under the third pivot zero.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 8/3 & -y & 8/3 \\ 0 & -4/3 & -y & -61/3 \\ 0 & 0 & 3y+3 & 199/9 \\ 0 & 0 & \frac{15y-294}{49} & -14,775 \end{bmatrix} = \begin{bmatrix} 1 & 8/3 & -y & 2.66 \\ 0 & -49/3 & -y & -20.33 \\ 0 & 0 & 3y+3 & 22.11 \\ 0 & 0 & \frac{-117y-735}{98} & -25.83 \end{bmatrix}$$

E_3 $E_2 \cdot E_1 \cdot A$

$\frac{-117y-735}{98} = 0$ $\boxed{y = 6.28}$ $\boxed{\alpha = \frac{8}{3}}$ $\boxed{m = -2}$

$\boxed{b = -11}$ $\boxed{k = \frac{34}{49}}$

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b) the uppe tri matrix is

$$\begin{bmatrix} 1 & 8/3 & -6.28 & 2.66 \\ 0 & -49/3 & -6.28 & -20.33 \\ 0 & 0 & 21.84 & 22.11 \\ 0 & 0 & 0 & -25.83 \end{bmatrix}$$

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Q3 $n \times n$ symetric matrix $H = I - 2uu^T$

a)

$\boxed{H^2 = I}$ since $\boxed{H = H^T}$

$$H^2 = (I - 2uu^T)^2 = (I - 2uu^T)(I - 2uu^T)$$

$$= I^2 - I\,2uu^T - 2uu^T \cdot I + 4uuu^T \cdot uu^T$$

$$= I\,(I - 2uu^T - 2uu^T) + 4u|u|^2 \cdot u^T$$

$$= I - 2uu^T - 2uu^T + 4u\|u\|^2 u^T$$

$\|u\|^2 = u^T \cdot u$

$|u| = \pm\sqrt{u \cdot u^T}$

As we know

$\|u\| = 1$

$$H^2 = I - 4uu^T + 4uu^T$$

$$= I - 4 \cdot (1) + 4 \cdot (1)$$

$\boxed{H^2 = I}$

b)

c) $H \cdot u = (I - 2uu^T)u$

$$= u - 2uu^T u$$

$$= u - 2u$$

$$= -u$$

$$\lambda = -1$$

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Q4) a)

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

$$\int dv = \int_0^1 (-v+w).dt \quad \text{so} \quad \underline{v = -v+w} \ \checkmark$$

$$\int dw = \int_0^1 (v-w).dt \quad \text{so} \quad \underline{w = v-w} \ \checkmark$$

b)

$$\frac{dv}{dt} = -v+w \qquad \frac{dw}{dt} = v-w$$

$$\begin{bmatrix} v_{t+1} \\ w_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix} \rightarrow \begin{bmatrix} v_1 \\ w_1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ w_1 \end{bmatrix} = \begin{bmatrix} +30 \\ -30 \end{bmatrix} \quad \text{at} \quad \boxed{t=1}$$

$$\begin{bmatrix} -30 \\ -30 \end{bmatrix} + \begin{bmatrix} 10 \\ 40 \end{bmatrix} \rightarrow \quad V \text{ tank}_1 \qquad \begin{matrix} 30 \rightarrow w \\ 10 \rightarrow v \end{matrix}$$

$$\frac{30}{40} \cdot 100 = \boxed{\%75} \text{ liquid}$$

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c)) singular matrix için

$$[A-\lambda I]=0 \rightarrow \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = (1-\lambda)^2-1=0$$

$$1-2\lambda+\lambda^2-1=0$$

$$\lambda^2-2\lambda=0$$

$$\lambda_1=0 \quad \lambda_2=2$$

at $\lambda_1=0$

$$\begin{bmatrix} 1-0 & -1 \\ -1 & 1-0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

at $\lambda_2=2$

$$\begin{bmatrix} 1-2 & -1 \\ -1 & 1-2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \cdot x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2=\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_k = S\Lambda^k S^{-1} u_0 \rightarrow u_k = S\Lambda^k S^{-1} C$$

$$u_k = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0^k & 0 \\ 0 & 2^k \end{bmatrix} C$$

$$u_0 = \begin{bmatrix} 10 \\ 40 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

x_1 x_2

$$c_1+c_2=10$$

$$c_1-c_2=40$$

$$\boxed{c_1=25}$$

$$\boxed{c_2=-15}$$

$$u_k = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0^k & 0 \\ 0 & 2^k \end{bmatrix} \begin{bmatrix} 25 \\ -15 \end{bmatrix}$$

for steady state $k \rightarrow \infty$

$$u_\infty = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & \infty \end{bmatrix} \begin{bmatrix} 25 \\ 15 \end{bmatrix}$$

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